



Spatiotemporal matching method for tracking pavement distress using high-frequency detection data

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Abstract

Various algorithms based on deep learning have achieved promising results in pavement distress detection. However, the detected distresses are not tracked throughout the life cycle. In long-term application scenarios, pavement distresses may take on different forms due to image acquisition mode, distress development, and environmental change, which make tracking distresses a tough question. We present in this study a spatiotemporal matching method based on high-frequency real pavement distress datasets. Pavement distresses of fixed routes were collected 30 times over 5 months, and distresses with spatiotemporal information were obtained at time series. We apply image rectification, stitching and distress class, and bounding box generation algorithms for pre-processing to align the collected images to the same-detail level and angle. A four-step spatiotemporal matching module is designed, including global positioning system (GPS) filtering, class filtering, relative position filtering, and distress feature filtering. The results reveal that the comprehensive rank-3 hit rate of the matching method reaches 88.73%, and the method is robust to environmental factors, which helps show performance decay of distresses and the effect of maintenance operations. It is concluded that the spatiotemporal matching method is convenient to operate, and it lays the foundation for an agency to track distress evolution and make timely treatment of distresses in the life cycle.

1 | INTRODUCTION

The occurrence and development of pavement distresses can shorten the service life of the road (Janoff et al., 1985; Smith et al., 1997) and affect driving safety (S. K. Chen et al., 2019). An early comprehensive inspection of distresses can bring sizable economic benefits (McLean & Foley, 1998). Automatic pavement distress evaluation has become mainstream and plays a critical role in the rapid assessment of road conditions and timely maintenance (Javadinasab

Hormozabad et al., 2021). With the development of computer vision technology, automated distress assessment has gradually focused on 2D images (Coenen & Golroo, 2017).

The existing acquisition techniques for pavement distress image data vary. The France road surface damaged dolly (GERPHO) system, first developed and put into use in the 1970s, realized synchronous acquisition of pavement images and location information (Chuo, 2009). In the 1980s, the Komatsu system in Japan applied video storage technology (Kim, 2008). In the mid-1990s, the generation of charge coupled device (CCD) imaging reduced

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equipment costs and improved image resolution. The representatives of this period were the Canada automatic road analyzer (ARAN) system, Australia Hawkeye 2000 system, and Sweden PAVUE system (Prasanna et al., 2012). Since the 2000s, digital cameras based on high-speed line scans (Zalama et al., 2014) and infrared thermal imaging (Du et al., 2017) have made a breakthrough. Eventually, integrated nondestructive automatic inspection systems brought convenience to realize high-precision and rapid inspection of distresses (Coenen & Golroo, 2017).

Apart from data collection, the algorithm for automatic distress detection and identification is also a research hotspot. Among them, computer vision-based automatic pavement distress detection has been a subject of high interest to the pavement engineering areas over the past decades (Gopalakrishnan, 2018). The increase in total infrastructure has put forward higher requirements for maintenance in recent years (Yano et al., 1998). Although the existing detection algorithms can realize effective distress detection after a single inspection, there is still a lack of analysis of the distresses after long-term and repeated detection (Liu et al., 2021). Tracking pavement distress after long-term detection is a re-identification problem (Zakira et al., 2021). There are already some mature algorithms that can solve the problem of cross-camera vehicle re-identification and person re-identification. In these re-identification scenarios, a targeted image (query set) is matched against a gallery set (representing the previously captured images in the surveillance camera network). However, due to the lack of pavement performance evolution data with the continuous period and enough space coverage (Koch et al., 2015), there is no related application in road management systems. In most cases, distress occurrence is sudden (Khraibani et al., 2010) and low-frequency detection has no timeliness, so it is difficult to observe the change process thoroughly. Therefore, high-frequency pavement distress data have become an essential premise for tracking pavement distress and the new data-driven maintenance mode. Besides, distress tracking at the individual level is of great significance for infrastructure management and its benefits can be reflected in the following two aspects:

1. It helps road administrators arrange large-scale maintenance work (Medury & Madanat, 2013). Road administrators need to know which distresses develop quickly and need to be remedied in time. Besides, predicting which distresses have the tendency of development through evolution analysis is useful to arrange remedial plans reasonably (Herabat & Songchitruksa, 2003). These plans include making decisions on maintenance means (Morcous & Lounis, 2005), determining reasonable maintenance time (Donev & Hoffmann, 2020), establishing a scientific matrix of pavement maintenance

scheme (Bosurgi & Trifiro, 2005), managing reasonable maintenance funds (Bhargava et al., 2017; L. Gao et al., 2012; Volovski et al., 2017), and posting evaluation of maintenance work (Nobakht et al., 2018).

2. It facilitates the post-evaluation of remedial plans. Long-term tracking of distresses can show the condition of distresses before and after maintenance, which is helpful to analyze the actual effect of distress maintenance. Furthermore, due to the limitation of maintenance conditions (labor, machinery, and materials), distresses may not be repaired all at once. Part of the distresses will have complex evolution over a long period of time. Analysis of the information change after a distress remedy can further assist the evaluation and decisions of follow-up maintenance plans.

Therefore, this paper proposes a spatiotemporal matching framework for distress tracking over a long time period based on high-frequency detection data, which means to be able to re-identify the same distress spot after being detected and maintained multiple times. There are two main characteristics of the method: (1) the long-term analysis is based on high-frequency data; (2) distresses tracking is at the individual level. Our spatiotemporal matching method for tracking pavement distress can help solve several core difficulties in the process of tracking payment distress:

1. Morphological change of distress: After long-term use, the pavement is constantly worn, resulting in various distress forms. The morphological characteristics of the distresses will change. Our method can adapt to major changes by improving the timeliness of the inspection data.
2. Parameter deviations: Sensor accuracy limits the traditional location method based on GPS information, and even state-of-art equipment has a measurement error within ± 5 m. There will be a large uncertainty in distress matching when only relying on GPS location. Besides, GPS may drift and the signal may not be well received at locations with covers (tunnels, trees, etc.). GPS sample data are based on time intervals, which means data collection density varies with respect to vehicle moving speed. Therefore, our method includes other parameters to help correct these deviations.
3. Unstable interpretability of distress features: In the actual application scenario, due to the influence of image shooting and environmental factors, the pavement distress features change a lot. Therefore, our method improves the interpretability of distress features by combining global feature analysis and local feature analysis.

Section 2 of this paper reviews the state-of-the-art literature on distress detection and feature extraction and



object-tracking problem. Section 3 explains the data preparation and pre-processing. Section 4 introduces the framework of the spatiotemporal matching method and its detailed model architecture. Section 5 verifies the proposed detection model based on high-frequency pavement distress data and investigates its feasibility. Section 6 discusses the evolution of distresses tracked by spatiotemporal matching and the applicability of the algorithm. Section 7 summarizes the study outcomes and generates the conclusion.

2 | LITERATURE REVIEW

2.1 | Pavement distress dataset construction methods

In order to analyze distresses, scholars established the construction method of pavement distress datasets. These datasets included top-down images (such as pavement image dataset (PID), and German asphalt pavement distress (GAPs); Eisenbach et al., 2017; Majidifard et al., 2020), unmanned aerial vehicle (UAV) images (Silva et al., 2020), smartphone images (Maeda et al., 2018), street view images (Maniat et al., 2021), ground penetrating radar (GPR) images (Tong, Gao, Han, & Wang, 2017), and 3D asphalt surface images from PaveVision3D system (D. J. Zhang et al., 2017). These datasets could be classified into image level (Gopalakrishnan, 2018), block level (Tong, Gao, & Zhang, 2017), and pixel level (Bang et al., 2019). However, the above datasets may not consider long-term repeated observations of the same road. The possible inconsistency of the data formats will bring a difficulty to the integration of different data types for the realization of long-term road distresses tracking (Liu et al., 2022). In order to accumulate long-term pavement performance evolution information, the US Department of Transportation Federal Highway Administration established the Federal Highway Administration's Long-term Pavement Performance Database (Gopalakrishnan et al., 2017), including seven modules: inventory, maintenance, monitoring (deflection, distress, and profile), rehabilitation, materials testing, traffic, and climatic. The data contained rich information to help analyze the impact of multiple factors on road performance changes. However, neither the updates of the indicators are highly frequent, nor the data granularity good enough to meet the long-term tracking on distress level. Our spatiotemporal matching method can help analyze the evolution of road performance from a more specific perspective.

2.2 | Pavement distress detection and feature extraction methods

There were two main types of image-based distress detection algorithms: model-based and model-free. For model-

based algorithms, the general idea was to improve the image quality through an image enhancement algorithm (Ying & Salari, 2010) and to use intensity-thresholding segmentation (Oliveira & Lobato Correia, 2009), edge detection (Nhat-Duc et al., 2018), wavelet transforms (Ying & Salari, 2010), or texture-analysis techniques to achieve distress extraction. Path search (Q. Li et al., 2010), f^* seed-growing (Q. Li et al., 2011) optimization methods were used to improve the accuracy. However, it was difficult for the traditional algorithms to make fast batch detection (Cao et al., 2014) due to their complexity, which led to inflexible adaptation to the changeability of the road environment. Deductive reasoning algorithms were developed afterward. Daniel and Preeja (2014) used a support vector machine (SVM) to achieve simple distress classification. Kapela et al. (2015) extracted directional gradient histogram (histogram of oriented gradient [HOG]) features, and Quintana et al. (2016) applied a local binary pattern to obtain distress features. R. Li et al. (2018) presented a unified and purely vision-based method denoted as defect detection and localization network for simultaneous concrete defect detection and localization.

While model-free algorithms were mainly based on data-driven approaches, these algorithms could increase abstraction levels and improve performance in large-scale parallel computing (Amezquita-Sanchez et al., 2020). They could realize feature extraction and distress identification by automatically training a large number of samples (Cha et al., 2017). Among them, convolutional neural networks (CNNs) could relatively cope with complex distresses (Ale et al., 2018) of large-scale object recognition without the need for a separate feature extraction (L. Zhang et al., 2016) because it could learn a large number of mapping relations between input and output without accurate mathematical expressions. In addition, deep learning has also provided a new solution for pavement distress detection and feature extraction since it has made great progress in many fields of civil engineering, such as construction cost estimation (Rafiei & Adeli, 2018), sale prices estimation (Rafiei & Adeli, 2016), earthquake early warning system (Rafiei & Adeli, 2017a), recommender systems based on collaborative filtering (Martins et al., 2020), real-time vehicle trajectory imputation (C. Zhao, Song, et al., 2022), direct trajectory planning (C. Zhao, Zhu, et al., 2022), and so on.

The key to the application of these methods in the field of road distress detection is to have an effective object detector module. Modern object detector is usually composed of three parts: a pre-trained backbone, a head to predict classes and bounding boxes of objects, and a neck to collect feature maps from different stages. The backbone part can be visual geometry group (VGG), ResNet,

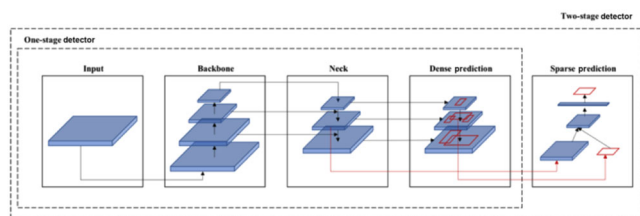


FIGURE 1 Comparison between one-stage and two-stage detectors (Bochkovskiy et al., 2020)

or Darknet53 (Bochkovskiy et al., 2020). The head part is usually categorized into two kinds, two-stages object detector and one-stage object detector. The most representative two-stage object detectors are region-based convolutional neural networks (R-CNN) series, such as deep convolutional neural networks (DCNN) (Yusof et al., 2018), Fast RCNN, and Faster RCNN (Cha et al., 2018; Ren et al., 2017; Song & Wang, 2019); while popular one-stage object detectors are you look only once (YOLO) series, single shot multibox detector (SSD) (Du et al., 2021), RetinaNet (Ale et al., 2018), and so on. For example, an end-to-end complete convolution network can realize automatic crack recognition at the pixel level (Yang et al., 2018). Besides, object detectors that can support multi-target recognition are also more suitable for spatiotemporal matching of pavement distress, which can detect multiple types of distresses. The neck is composed of several bottom-up paths and several top-down paths. Networks equipped with this mechanism include a feature pyramid network and a path aggregation network (Lin et al., 2017). Detector architecture is shown in Figure 1.

In the face of an insufficient pavement distress dataset, some scholars applied the transfer learning method to improve the performance of object detector models (Nie et al., 2018). For example, Y. Gao and Mosalam put forward a deep transfer learning framework for structural distress detection (Y. Gao & Mosalam, 2018). Li et al. proposed a transfer learning pipeline for untrained scenarios (Y. Li et al., 2021). Tong, Gao, Han, and Wang (2017) integrated CNN, Gaussian conditional random field, and uncertainty framework to detect pavement distresses. Some other scholars were devoted to generating datasets to help train models. K. Wang et al. (2017) and others used images in complex scenes to train networks, which greatly improved model generalization performance, and the detection accuracy was as high as 96%. In case of insufficient observed data, Maeda et al. applied a generative adversarial network (GAN) to generate distress images that could be used as new training data to improve detection accuracy (Maeda et al., 2021). Besides, a new pavement crack detection method was proposed to improve the detection accuracy by combining 2D grayscale images and 3D laser scanning data (Du et al., 2022; Huang et al., 2014; Weng et al., 2022). Some other advanced models

were also proposed, such as functional brain network (Hua et al., 2019), parameter sharing-based deep network (Reyes & Ventura, 2019), deep support vector neural networks (Diaz-Vico et al., 2020), and so on.

To sum up, in terms of distress detection, model-free algorithms are easier to operate, while model-based algorithms are more interpretable. The spatiotemporal matching method in this paper aims to combine the advantages of the two and dynamically improve the model performance to overcome the difficulty of distress feature extraction, which is the core step of long-term tracking.

2.3 | Object tracking and re-identification methods

Object tracking has become a research hotspot in the field of computer vision. Object tracking mainly includes three steps: data input, feature extraction, and metric learning. Among them, feature extraction is the most important for tracking algorithms. However, it is difficult for tracking due to illumination variations or appearance variations of the tracked objects.

To address these challenges, scholars designed several methods to extract features. There were three main categories: hand-crafted, machine learning-based, and deep learning. Hand-crafted features included gray features (Tian et al., 2021), color distribution (D. Wang et al., 2013), texture feature (Bousetouane et al., 2013), and HOG feature (S. B. Zhang & Wang, 2013). However, because these methods employed multiple manually designed features to represent the tracked target, the effect was not good enough in practical application (Bertinetto et al., 2016).

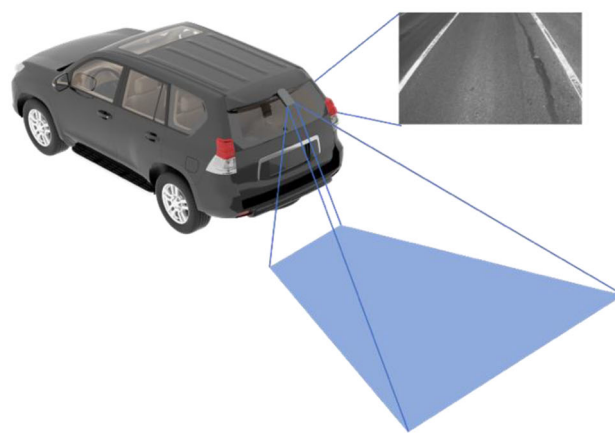
Machine learning-based methods mainly used speeded-up robust features (B. F. Zhang et al., 2014), scale-invariant feature transform (SIFT; Lowe, 2004), and HOG (S. B. Zhang & Wang, 2013). After feature extraction, different classifiers were applied such as linear regression, k-nearest neighbor (Guo et al., 2003), logistic regression, SVM (Brereton & Lloyd, 2010), Bayesian classification (X. X. He et al., 2019), and decision tree (Zurn et al., 2007).

Recently, deep learning has achieved great success in visual tracking tasks, particularly in single-object tracking. They have become the substitutions of the above manually designed features. Experiments showed that low-level features contained more target location information, while high-level features contained more semantic information, which was more robust than low-level features (Ma et al., 2015). There were two ways to obtain deep learning features-based tracker. The first was to transform the pre-trained deep neural network into a learning-based tracker. For example, GoogLeNet (Szegedy et al., 2015), VGGNet (H. Z. Chen & Fan, 2017), AlexNet (Krizhevsky et al., 2017), and ResNet (K. M. He et al., 2016) were used.

According to Z. D. Chen et al. (2020), the tracking problem was regarded as a parallel classification and regression problem, and a pre-trained classification network was used to solve the former. On the contrary, another method directly constructed the deep neural network. Z. P. Zhang et al. (2019) found that the network stride, receptive field, and spatial size of network output were very important for building a deep Siamese network. Then, they proposed cropping-inside residual (CIR) units, downsampling CIR unit, CIR-Inception unit, and CIRNeXt unit to optimize network performance. In addition, some new frameworks, such as generative countermeasure network (GAN) and attention mechanism, were applied to the depth tracker to effectively learn features. Also, in fine-grained visual recognition-based re-identification methods, local region features were extracted from different points. After features extraction, Euclidean distance matrix and cosine distance matrix were often used to measure the similarity between two targets.

Object tracking is a matching problem at a micro level, while re-identification is a matching problem at a macro level. The object-tracking problem only requires two adjacent data matches, but the re-identification problem requires a search from all datasets. Therefore, the re-identification problem is more complex, similar to pavement distress tracking. Vision-based re-identification methods mainly include five steps: data input, bounding box generation, training data annotation, model training, and image retrieval (Zakira et al., 2021). For the pavement distress tracking scenario, the objectives of these five steps are as follows:

1. Data input: Images of time series are key input. Mostly, these images are collected from an unconstrained environment (X. G. Wang, 2013).
2. Bounding box generation: Large-scale dataset is not conducive to analysis, so the region of interest is usually extracted first. Detection techniques are generally used to obtain the bounding box and class of the object (Sochor et al., 2019).
3. Training data annotation: The main purpose of training data annotation is to obtain the ground truth of distress identification (ID) so as to facilitate the calculation of ID loss during training.
4. Feature extraction and model training: The task of model training is to learn the weights of the feature extraction network and measure the bias according to the training dataset (Zhou et al., 2020).
5. Image retrieval: Image retrieval is a task of matching targeted distress (query image) over a gallery set (Y. Z. Zhao et al., 2020).



(a) Schematic diagram of acquisition equipment layout and field of view

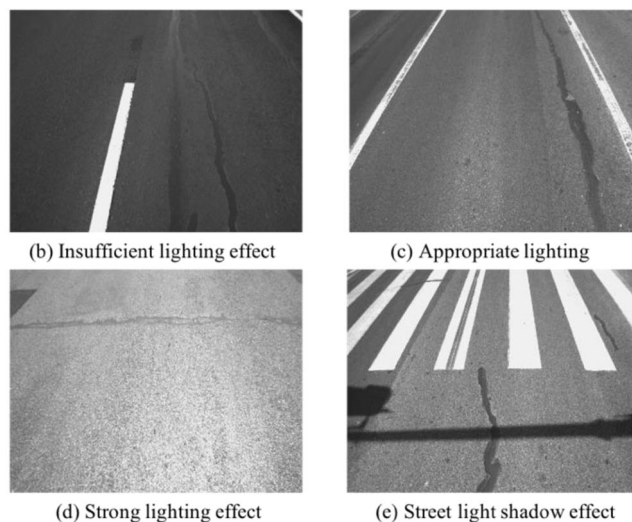


FIGURE 2 (a) Schematic diagram of acquisition equipment layout and field of view; and examples (b), (c), (d), (e) of the dataset with various environmental conditions

3 | DATA PREPARATION AND PRE-PROCESSING

3.1 | Image collection

A series of images were obtained from Shanghai Expressway Network. The leading acquisition equipment consisted of an industrial camera, GPS-integrated navigation receiver (GPS module and inertial navigation module), and data storage industrial computer. It could effectively collect road images with corresponding longitude and latitude coordinates, timestamps, and vehicle direction angles.

During the acquisition process, the industrial camera was placed on the vehicle's rear windshield at an angle of 45 degrees to the ground to capture road images. The position of the industrial camera and visual field are shown in

Figure 2a. Furthermore, the distance between the camera and the pavement surface was about 2 m. The vehicle and the camera angle remained unchanged during data collection. The GPS-integrated navigation receiver was installed on the roof and adsorbed by a magnet.

Pavement images of a fixed route were collected 30 times over 5 months (from June to October 2021). The images collected each time covered a 20 km single-vehicle road surface. The resolution of the images was 1920×1080 pixels. We dynamically intercepted the images according to the vehicle speed in the acquisition state using the variable frame rate method (Yuan et al., 2004). The number of frames captured per second was set as Equation (1):

$$f = \left\lceil \frac{v}{2.4l} \right\rceil + 1 \quad (1)$$

where f (pic/s) was the number of frames captured per second, v (km/h) was the driving speed, and l (m) was the length of the road in a single image.

A total of 180,000 road images and spatiotemporal information were obtained, occupying about 105 GB of storage space. The accuracy of the proposed spatiotemporal matching algorithm was verified on this dataset. Also, the dataset included a variety of environmental conditions as shown in Figure 2b–e. It could also help verify the robustness of the method.

The collected dataset contained seven distress classes as shown in Figure 3. They were cracking (longitudinal cracking and transverse cracking), alligator cracking, pothole, patched cracking, patched alligator cracking, patched pothole, and hinged joint.

In order to make the collected data better support the subsequent spatiotemporal matching algorithm, a series of pre-processing is carried out in advance. The pre-processing contains: (1) image rectification, (2) image stitching, (3) distress class and bounding box generation as shown in Figure 4.

3.2 | Image rectification

There is no distortion in the equipment used for data acquisition in this paper. However, there exists an inclined perspective deformation due to the camera mounting angle. This deformation will lead to significant differences in distress sizes at different locations of the images, so the image rectification algorithm is used for correction.

A 3×3 transformation matrix is used to correct the original image to a new plane. The transformation matrix under the same acquisition condition is unique as shown

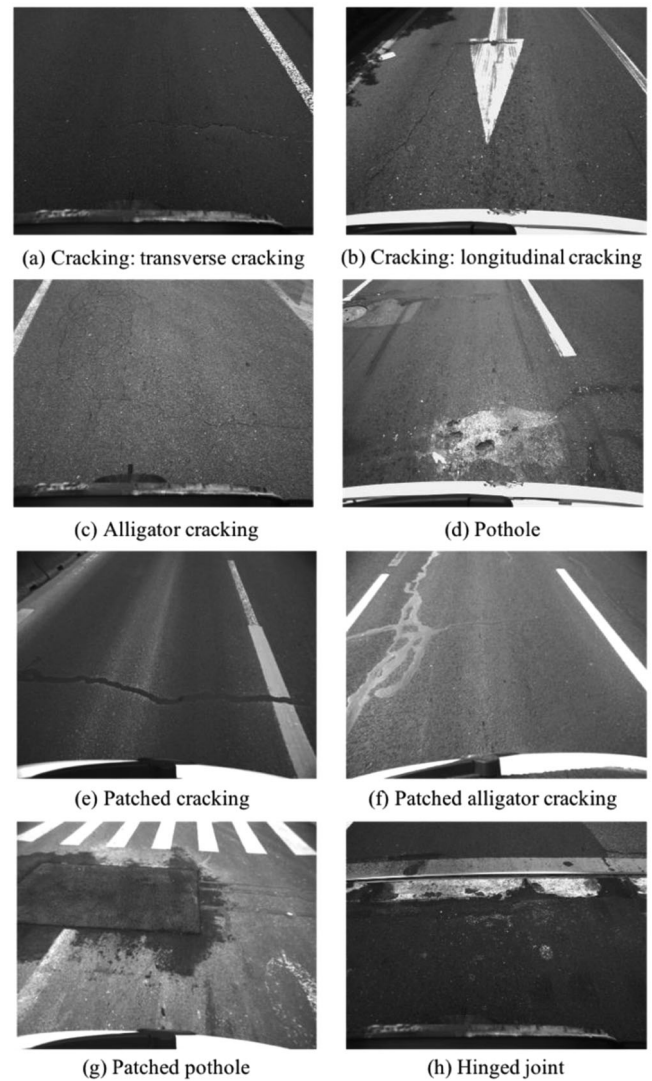


FIGURE 3 Seven classifications of pavement distresses

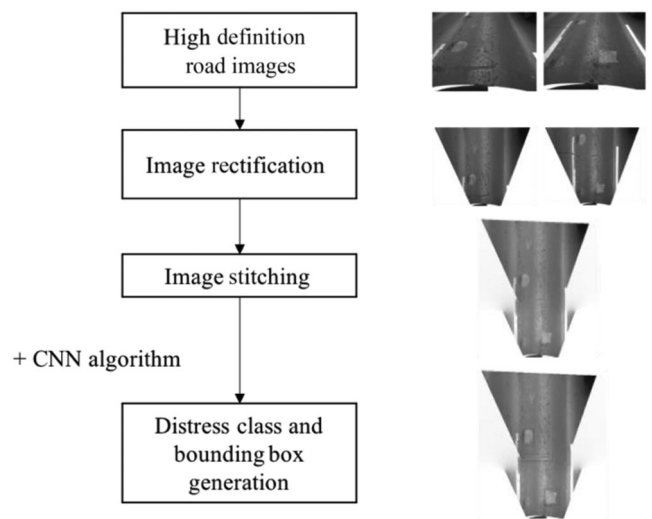


FIGURE 4 Flow chart of image pre-processing. CNN, convolutional neural network



TABLE 1 Comparison of different algorithms on pavement distress identification performance (adapted from Du et al., 2021), with permission)

Types of pavement distress (PD)	YOLOv3 (batch = 64)		YOLOv3 (batch = 96)		Faster R-CNN		SSD	
	Average precision (AP)	F1 score	AP	F1 score	AP	F1 score	AP	F1 score
Cracking	48.5%	0.4375	45.8%	0.4567	49.3%	0.5488	35.7%	0.3467
Patched cracking	69.8%	0.6673	69.2%	0.7126	72.3%	0.7658	56.0%	0.5721
Pothole	38.6%	0.3631	38.3%	0.4088	34.9%	0.3356	33.8%	0.3278
Patched pothole	63.3%	0.6855	61.6%	0.6579	62.7%	0.6753	55.1%	0.5689
Alligator cracking	50.6%	0.5134	48.6%	0.5076	53.5%	0.5690	46.6%	0.4875
Patched alligator cracking	41.8%	0.4289	40.9%	0.4055	41.9%	0.4432	34.4%	0.3567
Hinged joint	94.3%	0.9534	93.2%	0.9496	90.0%	0.9135	76.9%	0.7801
Operation Time/s	364.965		366.415		4359.300		496.696	

in Equation (2):

$$[x', y', w'] = [u, v, w] \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (2)$$

where u, v are pixel coordinates of the original image; w is the homogeneous coordinate; w' is the homogeneous coordinate after transformation; x', y' are physical coordinates of the transformed image; and a_{ij} refers to the transformation coefficient. Finally, images are corrected to the top view angle.

3.3 | Image stitching

Image stitching can effectively avoid the incomplete presence of large-scale distresses that occur in two adjacent images so as to improve the accuracy of subsequent matching of such distresses.

A fine-grained feature-matching and image-stitching method are used to eliminate duplications and visually demonstrate local pavement distress (Du et al., 2020). SIFT and random sample consensus (RANSAC) algorithms (M. F. Zhao et al., 2017) are used to generate region-wide images of roads with a specific length to effectively remove the repeated areas in the continuous acquisition and facilitate the spatial location determination of pavement distress. The SIFT operator is used to extract image features, which are invariant to image rotation change, scale change, and illumination change and stable to affine transformation, angle transformation, and noise. The SIFT operator is as follows:

1. Scale space extremum detection: This step searches image position on all scales. The Gaussian scale space of an image can be obtained by the convolution of its

image and Gaussian function as shown in Equation (3):

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y) \\ = \frac{1}{2\pi\sigma^2} e^{-\frac{(x-m/2)^2 + (y-n/2)^2}{2\sigma^2}} * I(x, y) \quad (3)$$

where the scale space $L(x, y, \sigma)$ is defined as a Gaussian function $G(x, y, \sigma)$ and $I(x, y)$ of the original image. m, n denote the dimension of the Gaussian template, and (x, y) is the pixel position of the image. σ is the scale factor. The large scale corresponds to the general features of the image, and the small scale corresponds to the detailed features of the image.

A more efficient Gaussian difference operator is used to replace the Laplace operator for extreme value detection as shown in Equation (4):

$$D(x, y, \sigma) = (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \\ = L(x, y, k\sigma) - L(x, y, \sigma) \quad (4)$$

2. Key point positioning: In each candidate position, the location and scale of the key points are accurately determined by fitting the 3D quadratic function. Besides, the low contrast key points and unstable edge response points are removed, and the matching stability is enhanced, and the anti-noise ability is improved.
3. Direction determination: One or more directions are assigned to each key point position based on the local gradient direction of the image. All subsequent operations on image data are related to the direction, scale, and position of key points so as to provide invariance for these transformations.
4. Key point description: In the neighborhood around each key point, the local gradient of the image on the

selected scale is measured. These gradients allow large local shape deformations and illumination variations.

In the process of SIFT feature extraction, each feature point has a corresponding feature vector, and the matching degree of any two feature points is judged according to the similarity between the feature vectors. The matching vector is constrained by the following Equation (5) to improve the matching accuracy of the feature vector:

$$\left\{ P_R \left| d(P_R, R) = d_{\min}, \frac{d_2(P_i, R)}{d(P_R, R)} \right\} t_d \right\} \quad (5)$$

Where P_R is the closest feature point to R , P_i is the second closest feature point to R , d refers to the Euclidean distance between the two features, and t_d is the distance threshold. The significance of the above equation is that only when this pair of matching has significant advantages over other matching, is the matching retained. When the distance between a feature point and many feature points is very small or close, it is considered that the feature point does not match other feature points. In order to further improve the matching accuracy and eliminate the wrong matching, a RANSAC (M. F. Zhao et al., 2017) algorithm is used to optimize and eliminate the wrong matching points.

The key to fuse multiple images is to find the overlap between the two images. Image fusion produces a consistent or smooth transition effect in terms of lighting effect and object shape. A fusion algorithm based on a dynamic weighted average is designed, and its weight is adjusted dynamically according to the bias of overlapping area as shown in Equation (6):

$$\begin{cases} F(x, y) = \lambda_A A(x, y) + \lambda_B B(x, y) \\ \lambda_A = 1 - \frac{x - x_{\min}}{x_{\max} - x_{\min}} \\ \lambda_B = 1 - \lambda_A \end{cases} \quad (6)$$

where $F(x, y)$ represents the fused image, $A(x, y)$ and $B(x, y)$ represent any two images participating in the fusion, λ_A and λ_B refer to the weight of the two images, x is the coordinates of the current pixel to be fused in the horizontal direction, and $x_{\min}$ and $x_{\max}$ are the beginning and end positions of horizontal overlap.

3.4 | Distress class and bounding box generation

In order to obtain important parameters in spatiotemporal matching, it is necessary to use the CNN algorithm to obtain the specific class and the bounding box coordinates of distresses.

TABLE 2 List of a priori parameters and related spatiotemporal matching steps

Parameter name	Access to parameters	Related spatiotemporal matching steps
GPS coordinates	Directly obtained by acquisition equipment	1. GPS filtering
Distress class	1. Image rectification; 2. image stitching; 3. distress class and bounding box generation	2. Class filtering
Bounding box coordinates		3. Relative position filtering

After preliminary research (Du et al., 2021; Ren et al., 2017), it is found that the detector in one stage has a good effect on pavement distress identification. In order to choose the best model framework for pavement distress detection and classification, comparative experiments were carried out (Du et al., 2021). All of the training and testing tasks were performed on a workstation with four graphics processing units (GPUs) (Nvidia Geforce Titan X × 4ea). The effect and efficiency of different algorithms were compared. Also, the effects of different ambient lighting on model performance were compared and analyzed. The detailed network structure and results can be seen in the previous research paper as shown in Table 1.

In this study, the YOLOv3 model is chosen as our framework to detect pavement distresses. The reasons are as follows:

1. First, research shows that feature semantic information of a low level is relatively small, but the target location information is accurate; feature semantic information of a high level is rich, but the target location information is relatively rough. YOLOv3 adds multi-scale recognition capabilities to combine both advantages. YOLOv3 introduces an up-sample structure, and its detection layer is fused with three-level feature layers. This method has advantages in feature extraction and discrimination of small objects and is suitable for the detection of pavement distress.
2. Second, in order to meet the needs of tracking pavement distress, it is necessary to collect a large amount of image data on the road surface. As a calculation, the storage needs approximately 300 pic/km*0.6 MB/pic = 0.18GB/km. YOLOv3 is faster than other deep learning frameworks that can save plenty of time.

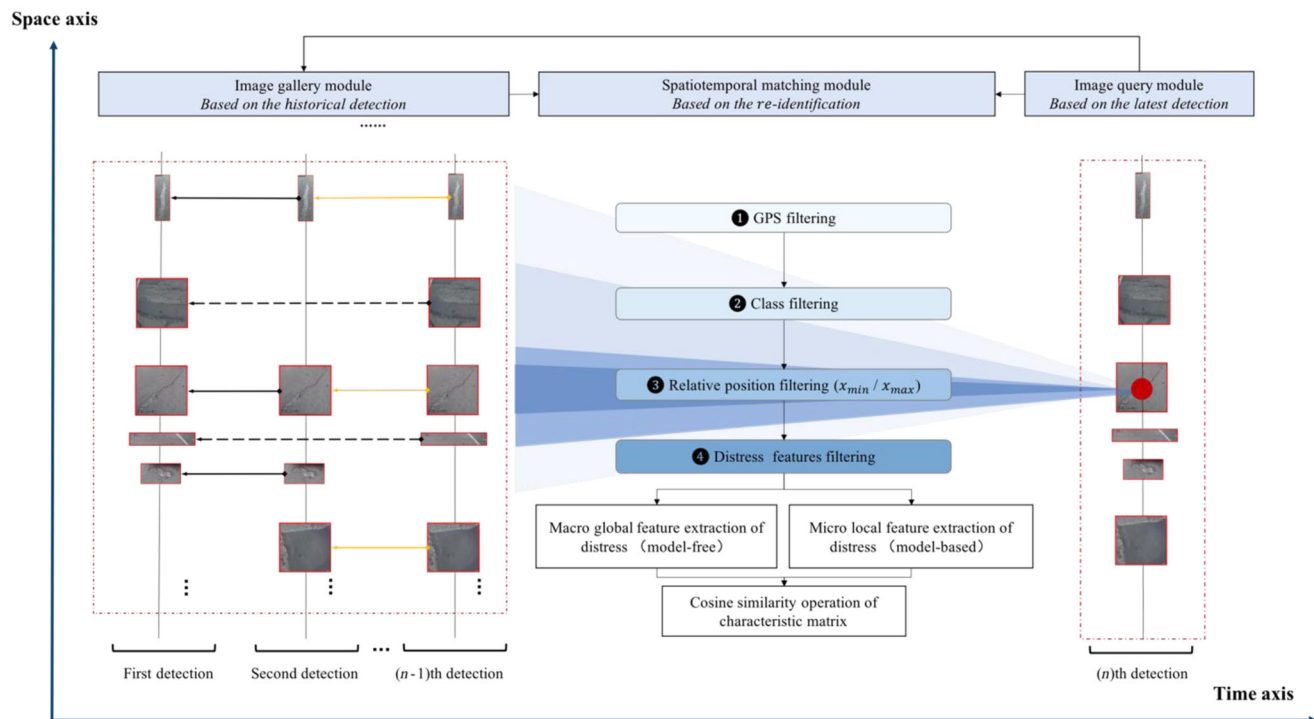


FIGURE 5 Overall pipeline of spatiotemporal matching method

3. Finally, YOLOv3 framework is easy to operate. The whole detection pipeline is an end-to-end framework. It is highly robust to complex environments, and it is suitable for continuous analysis of pavement distress images collected at high frequencies.

After pre-processing, important parameters can be obtained, which are the prior parameters input of the spatiotemporal matching method for tracking pavement distress. Table 2 summarizes these parameters. These parameters are used for the region of interest filtering process in spatiotemporal matching.

4 | METHODOLOGY

4.1 | Spatiotemporal matching framework

The state-of-art GPS equipment used for data collection has a measurement error within ± 5 m. There will be a large uncertainty in distress matching when only relying on GPS location. Besides, considering GPS drift and signal loss, it is necessary to supplement other features. Therefore, a spatiotemporal matching method for tracking pavement distress is proposed based on high-frequency detection data. The overall pipeline is shown in Figure 5.

A four-step spatiotemporal matching module is designed. Taking the latest detection data as the query module and the previous data as the gallery module, spatiotemporal matching based on re-identification can be carried out. It includes the following four steps:

1. GPS filtering: The first filtering is realized by using the longitude and latitude coordinates of GPS. According to the sampling error of GPS, the filtering range of GPS is set.
2. Class filtering: according to the distress class in the detection results of CNN, distresses of the same category in the range can be extracted.
3. Relative position filtering: According to the $x_{\min} / x_{\max}$ of bounding box, the relative location of distress from the left and right lane lines can be easily obtained. It is helpful for the preliminary screening of data samples.
4. Distress feature filtering: Feature extraction and model training are separately processed based on macro-global feature extraction of distress (model-free) and micro-local feature extraction of distress (model-based). In this way, the advantages of global feature analysis and local feature analysis can be combined.

Finally, the characteristic matrices of the two methods are obtained. By calculating the cosine similarity, the distress with the highest similarity is selected to complete the matching. Through multiple matching over a long

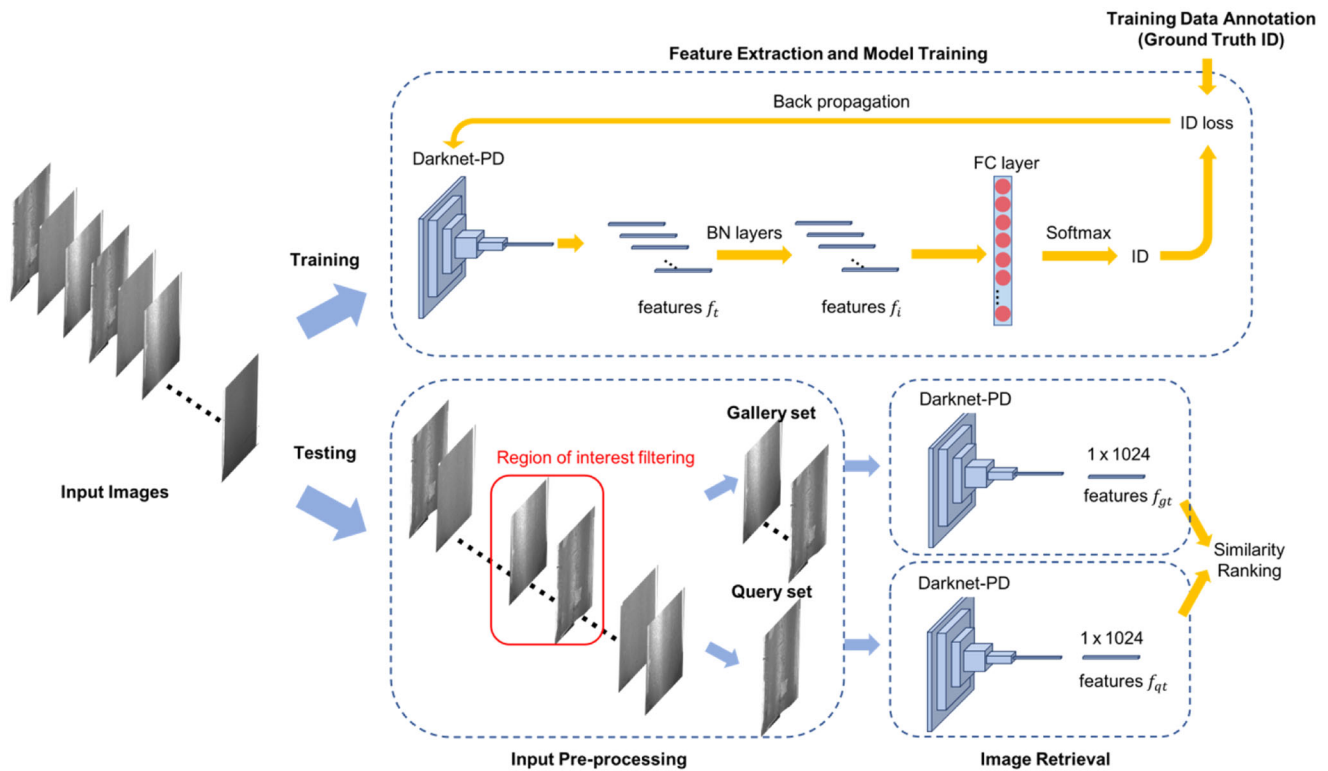


FIGURE 6 Novel deep learning framework for pavement distress re-identification based on macro-global feature extraction. BN, batch normalization

period of time, the continuous tracking of the same distress can be realized. It is worth noting that the first three steps are to reduce the candidate region by using a priori parameters obtained by data pre-processing. It has the advantages of improving performance and saving computing time.

In addition, the upsampling method is used to solve the problem of uneven positive and negative samples. A batch of negative samples is randomly copied to join the training to achieve the balance with the number of positive samples.

4.2 | Pavement distress re-identification based on macro-global feature extraction

We have established a novel deep learning framework for pavement distress re-identification (as shown in Figure 6), called Darknet pavement distress (Darknet-PD). The framework consists of five parts. First, the pre-processed images are as data input and are divided into a training set and testing set according to random sampling.

For the training set, the training data annotation is completed by manual annotation in advance, and the ground truth IDs of the distresses are obtained. The feature extraction and model training modules are established for follow-up training. First, the training set goes through

Darknet-PD, and the feature vector is obtained. Second, we set the batch normalization layers (J. R. Wang et al., 2019) to make the setting of network hyperparameters more free, such as setting a larger learning rate and realizing more random network initialization. At the same time, the convergence speed of the network is faster and the performance is better. Third, we set up a 1000-dimension fully connected (FC) layer (Khan et al., 2021) to make it possible for multiple ID classifications of distresses; the dimensions of the FC layer can be dynamically adjusted according to the number of distresses required to be tracked. Fourth, the softmax activation function (Rafiei & Adeli, 2018) is used to generate the distress ID, and the ID loss is calculated by comparing it with the ground truth ID. Finally, the hyperparameters of the whole training model are optimized through the back propagation mechanism.

After the model training, the effect verification is carried out on the testing set. First, input pre-processing is set, and the region of interest is filtered through a priori parameters (GPS coordinates, distress class, and bounding box coordinates). The data to be matched are automatically divided into query sets, and other data in the region of interest are taken as the gallery set. Then, during the image retrieval section, the data of the gallery set and query set are put into the trained Darknet-PD model at the same time and generate 1×1024 dimension feature vectors. Then, through

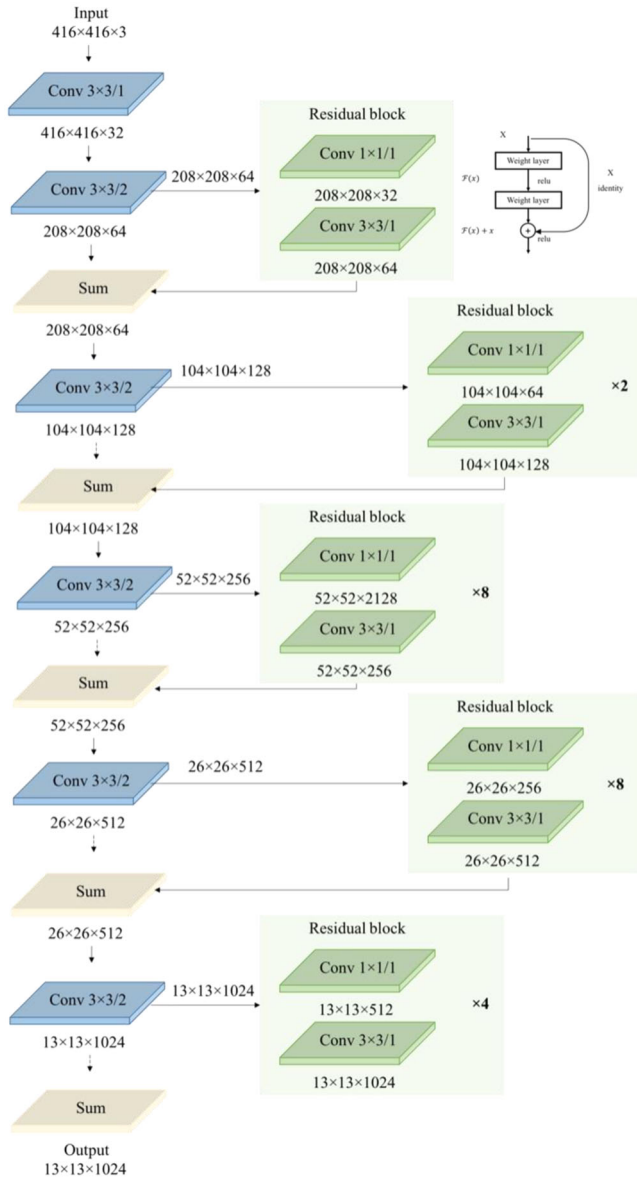


FIGURE 7 Darknet-PD architecture for global feature extraction

similarity ranking in f_{gt} and f_{qt} , the top k distresses that are most similar to query ones are extracted.

The main structural units of the Darknet-PD network are subsampling convolution and residual networks (as shown in Figure 7). Subsampling under the average method is used to make the image conform to the size of the display area and generate a thumbnail of the corresponding image. For an image I with the size of $P_x \times P_y$, downsampling of S times is performed to get a resolution image with $(\frac{P_x}{S}) \times (\frac{P_y}{S})$. The residual network has set up quick links between some layers to improve the operation efficiency based on ensuring accuracy. At last, five subsamplings and 23 residual networks are conducted.

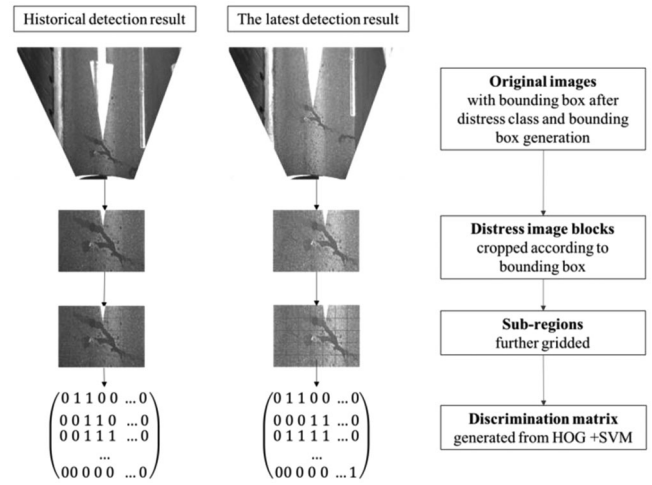


FIGURE 8 Model-based framework for pavement distress re-identification based on micro-local feature extraction

4.3 | Pavement distress re-identification based on micro-local feature extraction

Pavement distress re-identification based on macro-global feature extraction is a model-free method, but it may lack interpretability. In contrast, the model-based method is more explanatory. Therefore, this paper combines the advantages of the two and jointly carries out pavement distress re-identification. Model-based framework for pavement distress re-identification based on micro-local feature extraction is shown in Figure 8. The matching of two images is taken as an example.

First, the dataset is also divided into the latest detection results and historical detection results. Second, original images are processed by distress class and bounding box generation. Third, according to the bounding box, the original images are cropped into distress image blocks. Fourth, distress image blocks containing distress are further gridded into subregions. Finally, each subregion is put into a binary classification discriminant model for classifying whether there is a distress. The result of 0 represents that the subregion has no distress, and the result of 1 represents that the subregion has one distress. Then, a discrimination matrix of distress image blocks consisting of 0 and 1 can be generated.

The binary classification discriminant model is trained based on HOG + SVM (Figure 9). Red, green and blue (RGB) value of the input color image is converted to grayscale value. The color space of the input image is normalized by gamma correction. The gradient vector of each pixel in the image is calculated to capture the contour information. The gradient histogram of each cell is calculated to form the feature descriptor. The HOG descriptor of a block is obtained by concatenating the features of 2×2 cells in the block. The HOG descriptors of all blocks in the

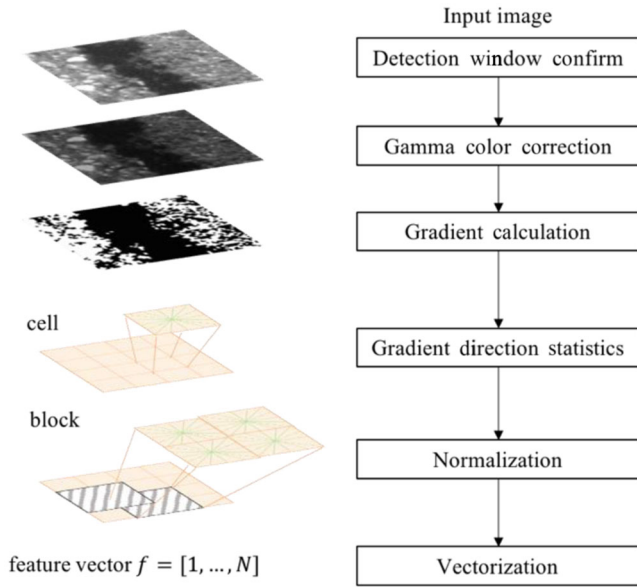


FIGURE 9 Flow chart of binary classification based on histogram of oriented gradient + support vector machine (SVM)

image are the final feature vector for binary classification. In general, gray, RGB, Hue-Luminance-Saturation (HLS), Hue-Saturation-Value (HSV), Luminance-Chrominance (YUV/ YCrCb) color modes are selected to extract HOG features.

In order to train the SVM classification model, the original image is cropped into $m \times n$ subregions by 64×64 pixels and labeled as two categories manually. A total of 3804 positive samples (1, subregion with distress) and 28,196 negative samples (0, subregion without distress) are obtained from the dataset. Furthermore, the HOG descriptor of each distress sample block is calculated. The dataset is divided into a training set and a test set with a ratio of 4:1. The YUV color channel is used to train the nonlinear SVM classifier based on the HOG feature, and the model is saved. The classification decision function is as shown in Equation (7):

$$f(x) = \text{sign} \left(\sum_{i=1}^N \alpha_i^* y_i \exp \left(-\frac{\|x - x_i\|^2}{2\sigma^2} \right) + b^* \right) \quad (7)$$

where $\alpha^* = (\alpha_1^*, \alpha_2^*, \dots, \alpha_N^*)^T$, $y_i \in \{0, 1\}$, $x_i \in R^n$, and $\exp(-\frac{\|x-x_i\|^2}{2\sigma^2})$ represents kernel function $K(x, z)$. The training samples and testing samples are shown in Figure 10. The accuracy of the algorithm is 88.172% in the training set and 88.438% in the testing set.

According to the discrimination matrix of two matching images, the cosine similarity is calculated to measure the difference. For two $m \times n$ dimensional matrices $\mathbf{A}$ and $\mathbf{B}$,

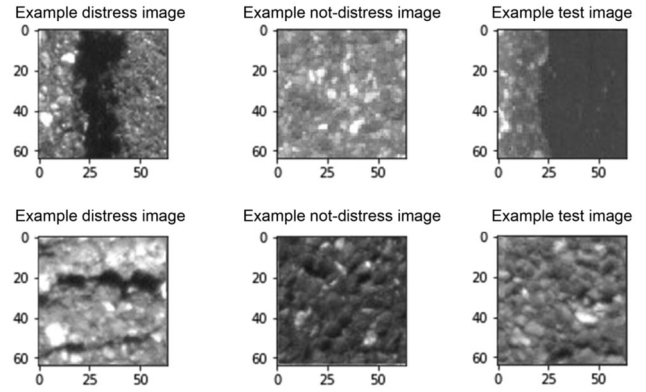


FIGURE 10 The training and testing samples of SVM model

the angle cosine is shown in Equation (8):

$$\begin{cases} \text{similarity}_c = \cos(\theta_c) = \frac{\mathbf{A}_c \cdot \mathbf{B}_c}{\|\mathbf{A}_c\| \cdot \|\mathbf{B}_c\|} = \frac{\sum_{i=1}^n \mathbf{a}_i \mathbf{b}_i}{\sum_{i=1}^n (\|\mathbf{a}_i\| \cdot \|\mathbf{b}_i\|)} \\ \text{similarity}_r = \cos(\theta_r) = \frac{\mathbf{A}_r \cdot \mathbf{B}_r}{\|\mathbf{A}_r\| \cdot \|\mathbf{B}_r\|} = \frac{\sum_{j=1}^m \mathbf{a}_j \mathbf{b}_j}{\sum_{j=1}^m (\|\mathbf{a}_j\| \cdot \|\mathbf{b}_j\|)} \\ \text{similarity} = \frac{n \cdot \text{similarity}_c + m \cdot \text{similarity}_r}{m+n} \end{cases} \quad (8)$$

where similarity_c and similarity_r are similarities of the column vector and row vector. The final similarity is the weighted average of them. θ_c and θ_r are cosine angle. $\mathbf{A}_c$, $\mathbf{B}_c$, $\mathbf{A}_r$ and $\mathbf{B}_r$ denote column and row vectors of matrices $\mathbf{A}$ and $\mathbf{B}$, respectively. $\mathbf{a}_i$ and $\mathbf{b}_i$ specify each column vector of matrices $\mathbf{A}$ and $\mathbf{B}$, while $\mathbf{a}_j$ and $\mathbf{b}_j$ specify each row vector.

Features of distresses extracted from macro global will generate matrix sequences with the size of 1×1024 , while features from micro local will result in the size of $m \times n$. Also, the elements in $m \times n$ dimensional matrix can be simplified into vector $\mathbf{A}'$ and $\mathbf{B}'$ of $1 \times (m+n)$ dimension.

The average cosine similarity of 1×1024 dimensional macroscopic characteristic matrix and $m \times n$ microcosmic matrix are calculated. When the above retrieval method is adopted, the image corresponding to the top k maximum cosine similarity ranking is taken as the successful matching image.

5 | RESULTS

This section implemented the algorithm testing. The above-mentioned dataset was randomly divided into training and testing datasets. The testing dataset contained 1938 images with 2099 distress detection results. According to manual data annotation, 948 groups of distresses were found, and the distresses at the same location had the same ID.

**TABLE 3** The rank-K hit rate results of the spatiotemporal matching (SM) method with various GPS filtering ranges and $k = 1, 2, 3$

Rank-K hit rate		$K = 1$	$K = 2$	$K = 3$	Operating time
No spills	GPS (3 m) + SM	79.71%	86.39%	88.73%	10'53"
	GPS (5 m) + SM	77.14%	82.68%	84.80%	11'32"
	GPS (10 m) + SM	78.59%	86.08%	88.44%	12'45"
	GPS (15 m) + SM	75.54%	83.75%	86.74%	14'15"
	GPS (30 m) + SM	69.83%	78.57%	82.52%	17'09"
With spills	GPS (3 m) + SM	71.01%	76.96%	79.05%	14'27"
	GPS (5 m) + SM	69.07%	74.62%	76.49%	15'02"
	GPS (10 m) + SM	63.37%	69.41%	71.32%	17'00"
	GPS (15 m) + SM	57.31%	63.54%	65.81%	19'01"
	GPS (30 m) + SM	47.46%	53.42%	56.11%	22'52"

After the region of interest filtering, query sets and gallery sets were set. For each image in one query set, similarities with all images in the corresponding gallery set were calculated. Rank-K hit rate was used as an evaluation index to show the accuracy rate of matching. The Rank-K hit rate was obtained by counting the proportion of the number of images with the same ID as the query image among the first k images sorted by similarity. The similarity of batch images are shown in Equations (9)–(11):

$$feature_{query} = [q_1, q_2, \dots, q_N] \quad (9)$$

$$feature_{gallery} = \begin{bmatrix} a_{11}, a_{12}, \dots, a_{1N} \\ a_{21}, a_{22}, \dots, a_{2N} \\ \vdots \\ a_{M1}, a_{M2}, \dots, a_{MN} \end{bmatrix} \quad (10)$$

$$S_M = \sqrt{\sum_{i=1}^N (q_e - a_{Me})^2} \quad (11)$$

where N stated the number of characteristic dimension, and $feature_{query}$ was the features extracted from every single query image. The gallery set contained M pictures, and the features of them were extracted as $feature_{gallery}$. q_e was a component of the extracted features of the query image, and a_{Me} was a component of the features of the M th image in the gallery set. S_M was the similarity of the query image and the M th image in the gallery set. The lower the S_M was, the greater the image similarity would be, with higher confidence that these two images were the same distress. S_M was sorted from lowest to highest. Then the rank-K hit rate was calculated according to the ground truth.

As shown in Table 3, the rank-K hit rate results of the spatiotemporal matching method with various GPS filtering ranges and $k = 1, 2, 3$ were explored. As mentioned

above, the accuracy of the GPS equipment used in this paper was 0.4 m, and the comprehensive positioning error is ± 5 m. In addition to seven classes of pavement distresses, the dataset also contained many spills. The performance of the model with and without spills was also analyzed.

Through the comparison of algorithm effects under different GPS filtering ranges, it could be found that:

1. With the expansion of the GPS filtering ranges, the performance of the algorithm decreased. However, there was a particular case (GPS(10 m) + SM, no spills) when the performance improved. Correct images increased when the GPS filtering range was expanded to about 10 m (around the GPS positioning error).
2. With the increase of K value, the effect of the algorithm was improved, which was consistent with common sense. When the number of images allowed to make mistakes increased, the probability of correct matching would increase.

The operating time of the method was short (between 10'53" and 22'52"). The computational efficiency can be helpful to implement frequent inspections and provide feedbacks.

Table 4 showed the performance comparison between the proposed spatiotemporal matching and GPS matching only. We could conclude that:

1. Adding distress feature filtering can improve the matching performance. Moreover, the smaller the GPS filtering range was, the more noticeable this improvement would be.
2. When the GPS filtering range was too large (e.g., 30 m), adding distress feature filtering had a negative effect on the spatiotemporal matching. When the candidate domain was too large, adding more distresses with similar characteristics might lead to a failure of the algorithm.

**TABLE 4** The rank-K hit rate comparison between the proposed method and GPS matching only

Rank-K hit rate		K = 1	K = 2	K = 3
No spills	GPS (3 m) + SM	79.71%	86.39%	88.73%
	GPS (3 m) only	58.43%	63.12%	64.99%
	GPS (5 m) + SM	77.14%	82.68%	84.80%
	GPS (5 m) only	69.13%	75.55%	77.97%
	GPS (10 m) + SM	78.59%	86.08%	88.44%
	GPS (10 m) only	73.69%	82.53%	85.15%
	GPS (15 m) + SM	75.54%	83.75%	86.74%
	GPS (15 m) only	74.10%	83.29%	86.05%
	GPS (30 m) + SM	69.83%	78.57%	82.52%
	GPS (30 m) only	74.10%	83.36%	86.19%
With spills	GPS (3 m) + SM	71.01%	76.96%	79.05%
	GPS (3 m) only	41.50%	44.74%	46.02%
	GPS (5 m) + SM	69.07%	74.62%	76.49%
	GPS (5 m) only	48.93%	53.36%	55.03%
	GPS (10 m) + SM	63.37%	69.41%	71.32%
	GPS (10 m) only	52.07%	58.17%	59.98%
	GPS (15 m) + SM	57.31%	63.54%	65.81%
	GPS (15 m) only	52.36%	58.69%	60.60%
	GPS (30 m) + SM	47.46%	53.42%	56.11%
	GPS (30 m) only	52.36%	58.74%	60.70%

3. GPS filtering range should be adjusted according to the positioning accuracy of GPS equipment.

Figure 11 showed the samples of successful matching results. The method could achieve spatiotemporal matching of most distresses and can meet engineering application needs. It could be seen that the algorithm has a sound effect, and it could effectively achieve the matching after multiple distress detections, especially for some prominent distress features.

However, there were still some false matches and missed matches, which were defined as matching errors. These situations were shown in Figure 12. There was inconsistent performance across different distresses. The leading causes of errors can be classified into the following four points:

1. Due to different illumination conditions, the image features before and after matching were too different.
2. The pixels in the distress area were unclear or the features were not apparent. For example, the matching of cracking might not be ideal because of their inconspicuous features.
3. The image shooting angles significantly differed, resulting in inconsistent features.
4. Uncontrollable lane change during the data acquisition process may result in data loss.

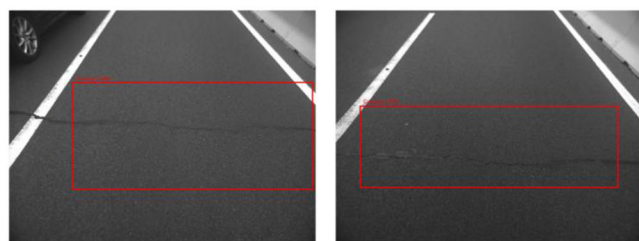
In the experiment, the acquisition time interval was 1 week. Since the inspections were of high frequency, and the distress change was minimal, the accuracy of the spatiotemporal matching method was relatively high. If the acquisition time interval expands, the distresses would have more apparent changes, and the accuracy of the spatiotemporal matching method might reduce. Thus, to receive a better matching performance, it is recommended to carry out high-frequency data acquisition. At present, many applications of portable detection equipment can achieve high-frequency detection.

6 | DISCUSSION

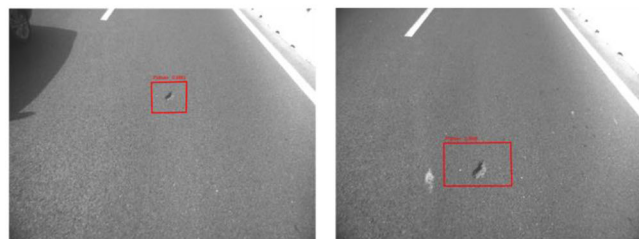
6.1 | Evolution of distresses tracked by spatiotemporal matching

The spatiotemporal matching method can effectively match the distress at time series and is helpful to track the evolution of the distresses. The typical cases of distress evolution found in the results of spatiotemporal matching are shown in Figure 13.

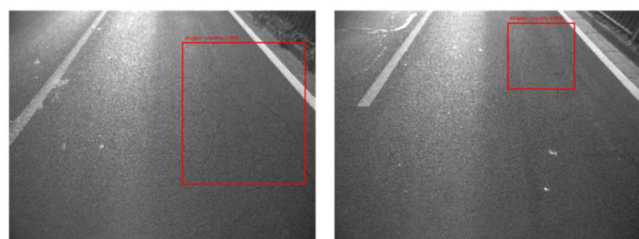
Figure 13a,b, respectively, shows that a transverse cracking and a pothole continued to develop during the detection cycle. Figure 13c shows that the hinged joint area



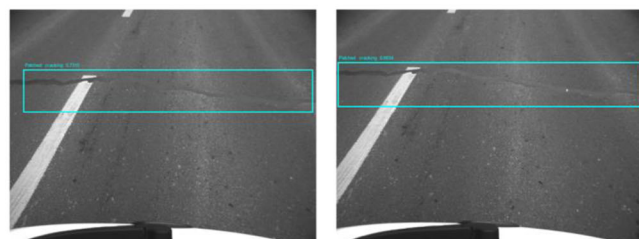
(a) Cracking



(b) Pothole



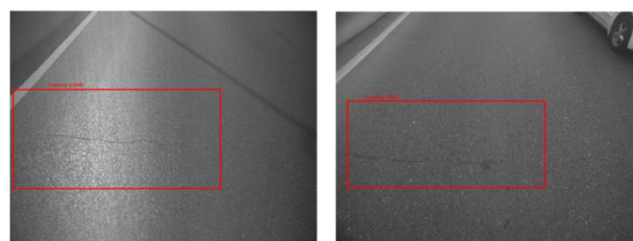
(c) Alligator cracking



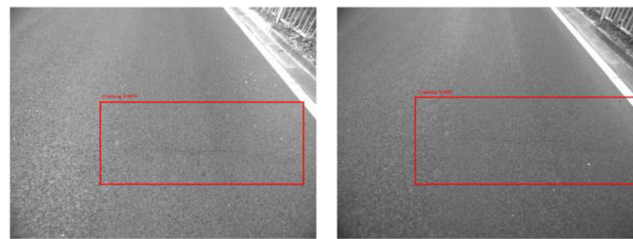
(d) Patched cracking

FIGURE 11 The samples of successful matching results of various types of distresses

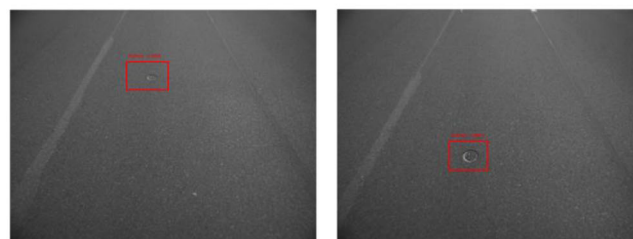
was worn but then was repaired. Figure 13d shows that transverse cracking reappeared after repair, and the crack width was increased. Figure 13e shows that the hinged joint area was worn, and then a repair block was made in the surrounding area. However, the repair was not ideal, and therefore the repair was made within the hinged joint area again. Figure 13f shows potholes around many repair blocks. After simple maintenance, the distress still increased. Finally, the whole section of the road was milled and repaved. Figure 13g shows that there were certain alligator cracking on the repaired crack area, accompanied by potholes. The state was maintained after repair, but the potholes and bulges were formed again, even with aggregates peeling. Finally, the whole pavement is milled.



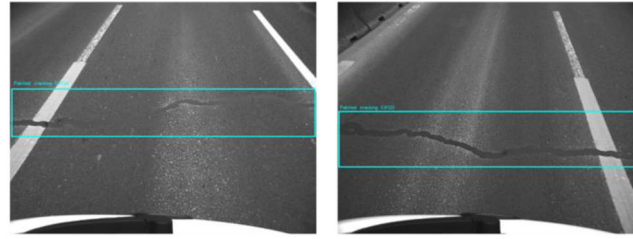
(a)



(b)



(c)



(d)

FIGURE 12 The samples of false matches or missed matches

It can also be seen that the evolution of general pavement distress has phased attributes, and the development roughly presents the stage from cracking to alligator cracking to pothole and then to raveling. Slight cracks often generate at first, and they are mainly fatigue cracks under dynamic load and reflection cracks under the influence of construction factors, water intrusion damage, temperature, and so on. Then, in the second stage, the cracks will gradually become wider. Sometimes, although the cracks are patched, the stiffness may be insufficient. When the load is too intense, they may develop into regional alligator cracks. In the third stage, alligator cracks are easily worn internally due to water infiltration, and then potholes are formed. In the fourth stage, under the influence of further hydrodynamic pressure, the aggregates will peel off, and the potholes will develop into pavement raveling.

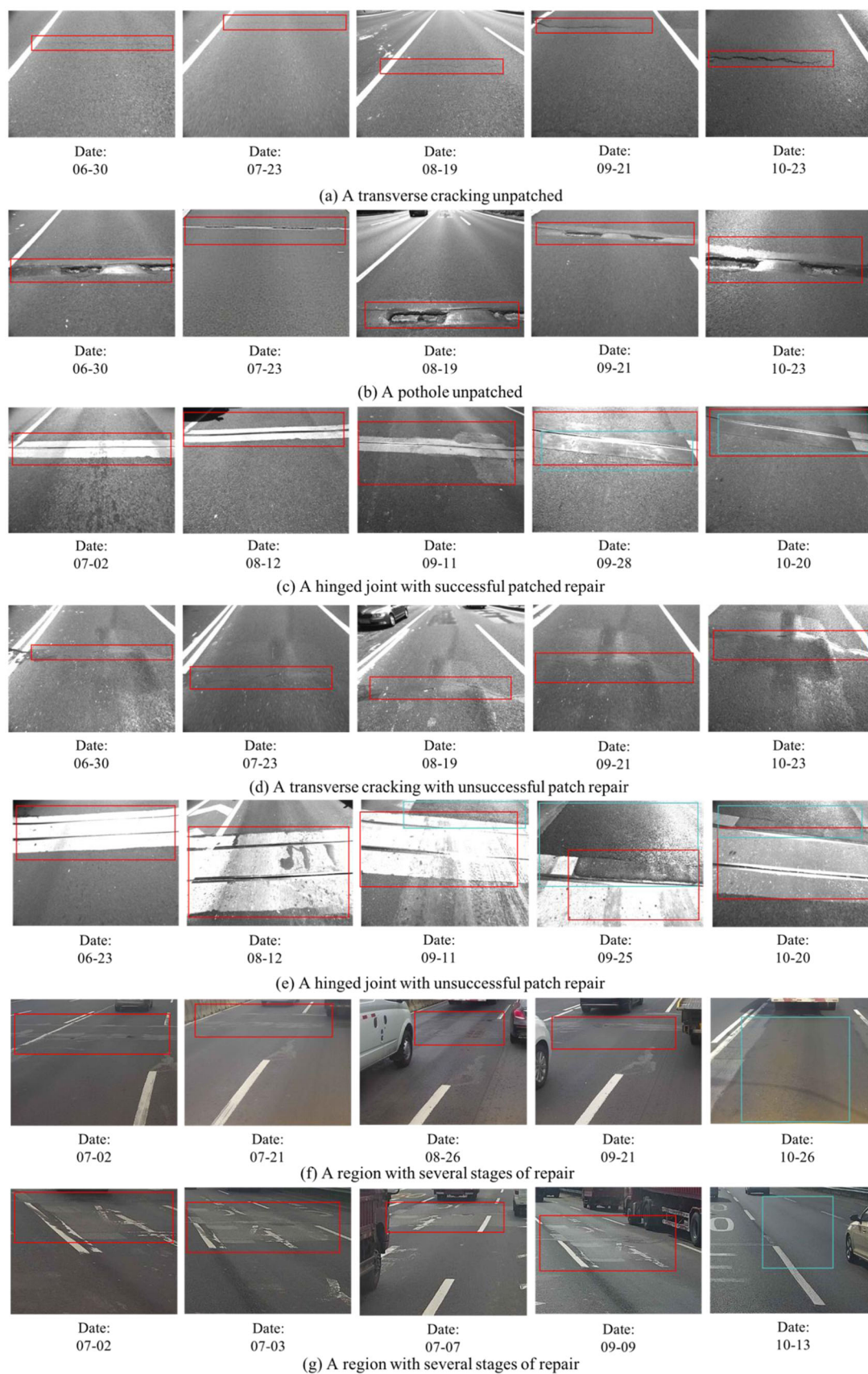


FIGURE 13 Typical cases of distress evolution found in the results of spatiotemporal matching

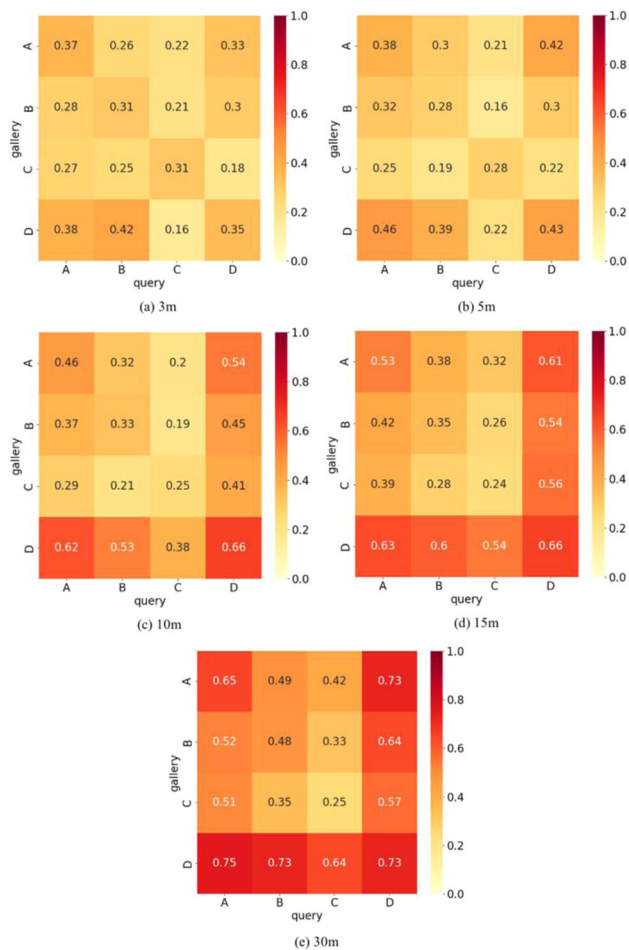


FIGURE 14 Comparison of the method error rate under different lighting conditions

6.2 | Applicability of spatiotemporal matching method

Figure 12 shows that the proposed algorithm is easily affected by external factors such as lighting conditions, image resolution, and image shooting angle and range. Therefore, if a large-scale engineering application is to be carried out, algorithm optimization is further needed to improve the robustness of the method.

The dataset in this study includes a variety of lighting conditions as shown in Figure 2b–e. Images are classified into four typical scenarios. It can effectively verify the robustness of the algorithm under different lighting conditions. In order to make it more specific, we compare and analyze the accuracy of cross-matching when the images under different lighting conditions are used as query sets and gallery sets. The results are shown in Figure 14.

In the figure, the horizontal axis represents the lighting condition of the query set, the vertical axis represents the lighting condition of the gallery set, and the A–D numbers

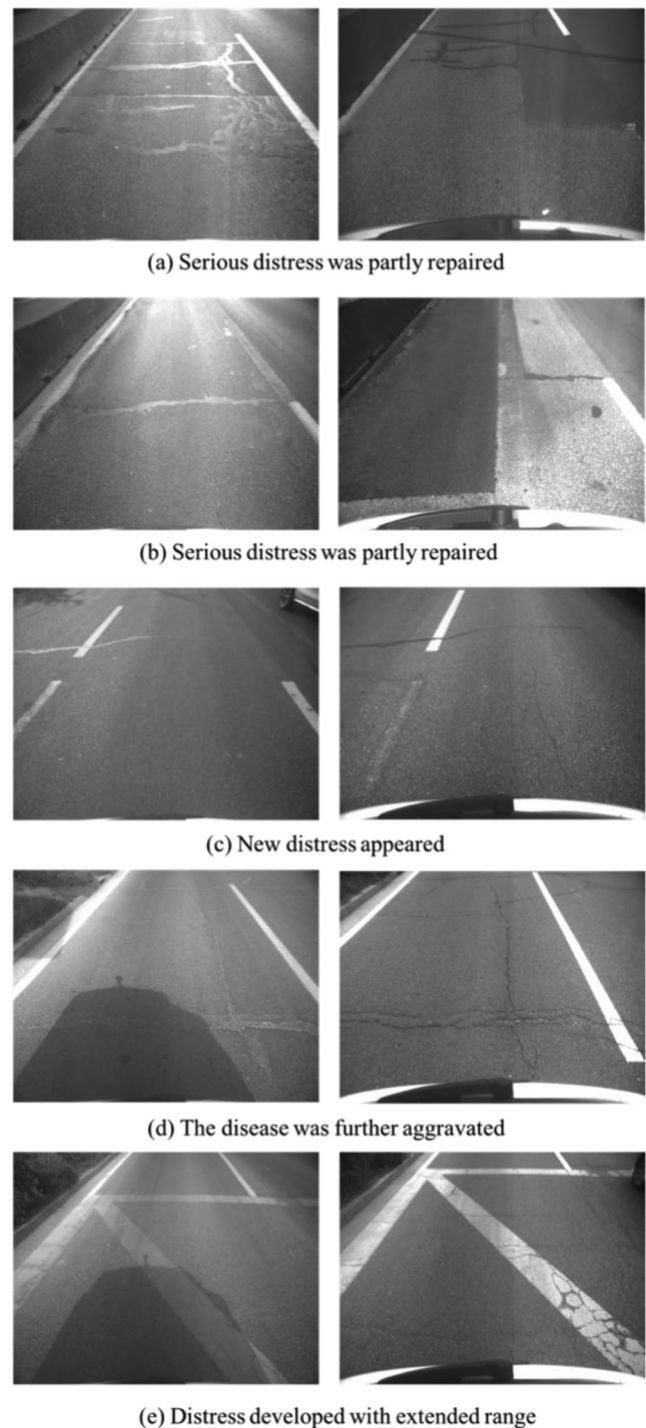


FIGURE 15 Situations of distresses with noticeable changes (left: historical detection, right: the latest detection)

respectively, represent: (A) insufficient lighting, (B) appropriate lighting, (C) strong lighting, (D) with shadows or occlusion. The results of cross-validation are shown. The numbers in the figures indicate the error rate of matching. The darker the color, the higher the error rate. On the contrary, the lighter the color, the higher the accuracy rate and the better the model performs.



A total of five cases are compared when the GPS filtering range is 3, 5, 10, 15, and 30 m. It can be concluded that with shadows or occlusion (D), the performance of the method is generally poor. When the illumination condition is appropriate (B) or strong (C), the performance of the method is better. Especially when the lighting is strong, the performance is the best. At the same time, we can also find that when different lighting conditions are divided into gallery set and query set, the influence weight on matching is equal.

Figure 15 also presents many situations when distresses have noticeable changes in the process of evolution. More factors need to be considered in this situation. Figure 15a,b shows that the distress was repaired by maintenance operation within the interval of two detections. Figure 15c shows that new distress was generated. Figure 15d,e shows that the extent of distress was aggravated or the scope of distress was expanded. In addition, the distress development cycle is different under various traffic volumes and climate conditions. The spatiotemporal matching of distresses under different environmental factors should be further studied and discussed.

7 | CONCLUSION

A spatiotemporal matching method is proposed for tracking pavement distress using high-frequency detection data, which can help track distress in the life cycle more efficiently.

At first, the data acquisition process lasted for 5 months and achieved regular pavement distress data on fixed lines. A total of 30 times of spatiotemporal data was obtained and a high-frequency real pavement distress dataset was established. We apply image rectification, stitching and distress class, and bounding box generation algorithms for pre-processing and obtain GPS coordinates, distress class, and bounding box coordinates. They are the prior parameters input of the spatiotemporal matching method for tracking pavement distress, which is used for the region of interest filtering process.

Second, a four-step spatiotemporal matching module is designed, containing GPS filtering, class filtering, relative position filtering, and distress feature filtering. Taking the latest detection data as the query module and the previous data as the gallery module, spatiotemporal matching based on re-identification can be carried out. We have evaluated the proposed method, and the comprehensive accuracy of distress tracking is 88.73% when $k = 3$ and GPS filtering range is within 3 m.

Then, typical cases of distress evolution found in the results of spatiotemporal matching are shown to prove the effectiveness of the algorithm. It can be concluded

that the method helps present performance decay of distresses and the effect of maintenance operations. The applicability and limitations of this method are also discussed. The error of this method mainly comes from the influence of external factors such as lighting conditions, image resolution, and image shooting angle and range. Also, the evolution of distress may cause difficulties in the analysis. The influence of illumination conditions on the matching method is quantitatively analyzed. The results show that the method is robust to environmental factors and lays the foundation for an agency to track distress evolution and make timely treatment of distresses in the life cycle.

For future work, distress evolution under different traffic volumes and climate conditions will be explored. The spatiotemporal matching strategy of distresses under different environmental factors shall be further studied and discussed. In addition, some novel supervised machine learning and classification algorithms can provide more reliable and fast prior parameters for spatiotemporal matching methods. The fusion of spatiotemporal matching methods and these algorithms can be the main direction of future expansion research. For example, dynamic ensemble learning algorithm (Alam et al., 2020) performs accurate classification by discovering the most effective feature space and finding the optimum number of features for accurate classification. Neural dynamic classification algorithm (Rafiei & Adeli, 2017b) has better generalization ability for the classification problem in complex road scenes. Finite element machine (Pereira et al., 2020) can support faster and more efficient machine learning algorithm operation and further improve model efficiency. These advanced algorithms can improve the efficiency and performance of the proposed spatiotemporal matching method.

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